

# **A Case Study of Colombian Engineering Students**

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## **Executive Summary**

This report analyzes the academic performance of 12,411 Colombian engineering students using data from *Data of Academic Performance Evolution for Engineering Students*. The analysis examines whether academic performance is associated with parental education, socioeconomic stratum, and gender.

The results indicate that socioeconomic stratum can be the strongest predictor of student performance, with gaps of 29.16 points between the highest and lowest strata. In addition, parental education is comparably influential, with gaps of 21.79 points (Father's education) and 25.93 points (Mother's education) between the highest and lowest education categories. In contrast, gender differences have minor effect on student performance.

Based on these findings, the report recommends establishing a first-year bridging program aimed at improving English proficiency for students from lower socioeconomic strata and conducting similar analyses in other disciplines to determine whether the same disparities appear.

### **1. Key Issues/Goals/Problems**

The Colombian Department of Education is interested in understanding whether academic performance among engineering students differs across different groups. Several factors may influence student outcomes, including parental education, socioeconomic background and gender. Understanding whether these factors are associated with student outcomes is essential for designing effective educational interventions. This report aims to

identify the factors and to provide data-driven recommendations for improving educational equity.

## **2. Decision Criteria**

Statistical significance was assessed at the conventional threshold of  $p < 0.05$ . For comparisons between two groups (gender), independent-samples t-tests were used. For comparisons across three or more groups (parental education, stratum, SISBEN), one-way ANOVA was used. The Global Score served as the primary outcome variable.

## **3. Data Analysis**

### **3.1 Descriptive Statistics**

This report analyzes a dataset, titled *Data of Academic Performance Evolution for Engineering Students*, developed by Professor Enrique De La Hoz and engineer Rohemi Zuluaga (De La Hoz). The data were obtained by cross-referencing databases from the Colombian Institute for the Evaluation of Education (ICFES). The dataset contains 12,411 engineering students from Colombia, of whom 5,043 (40.6%) were female and 7,368 (59.4%) were male. The dataset includes academic, social, and economic information collected at two stages in each student's educational experience: the final year of high school and the final year of the professional engineering program (the national standardized test for higher education). Academic variables include high school scores in Mathematics (MAT\_S11), Critical Reading (CR\_S11), Citizenship Competencies (CC\_S11), Biology (BIO\_S11), and English (ENG\_S11); and pro-level scores in Quantitative Reasoning (QR\_PRO), Critical Reading (CR\_PRO), Citizenship Competencies (CC\_PRO), English

(ENG\_PRO), Written Communication (WC\_PRO), and Formulation of Engineering Projects (FEP\_PRO). Social and economic variables include gender (GENDER), parental education (EDU\_MOTHER, EDU\_FATHER), parental occupation (OCC\_MOTHER, OCC\_FATHER), socioeconomic stratum (STRATUM), SISBEN classification (SISBEN) and household assets.

The analysis primarily focuses on seven outcome variables from the pro-level scores. Five modules are scored on a 0-100 scale: Quantitative Reasoning (QR\_PRO,  $M = 77.42$ ,  $SD = 22.67$ ), Critical Reading (CR\_PRO,  $M = 62.20$ ,  $SD = 27.67$ ), Citizenship Competencies (CC\_PRO,  $M = 59.19$ ,  $SD = 28.99$ ), English (ENG\_PRO,  $M = 67.50$ ,  $SD = 25.50$ ), and Written Communication (WC\_PRO,  $M = 53.70$ ,  $SD = 30.00$ ). One module, Formulation of Engineering Projects (FEP\_PRO), is scored on a 0-300 scale ( $M = 145.48$ ,  $SD = 40.13$ ). The Global Score (G\_SC), a composite measure of overall performance, is used as the primary outcome of interest ( $M = 162.71$ ,  $SD = 23.11$ ). All seven outcome variables contained complete data with no missing values. Table A1 in Appendix A presents detailed descriptive statistics. The main independent variables are gender (GENDER), parental education (EDU\_MOTHER, EDU\_FATHER), stratum (STRATUM), and SISBEN classification (SISBEN).

### **3.2 Parental Education and Academic Disparities**

A one-way ANOVA was conducted to examine whether father's education level was associated with student performance on the Global Score. Education level was originally recorded across 12 categories, including incomplete and unspecified levels. For this analysis, we kept six categories: Ninguno, complete primary, complete secondary, complete technical or technological, complete professional, and postgraduate education.

Incomplete and unspecified categories were excluded because they might obscure the progression. The result was statistically significant,  $F(5, 9079) = 148.28, p < .001$ , indicating that the Global Score differed substantially across father's education levels. Mean of the Global Score increased from complete primary education ( $M = 155.20$ ), complete secondary ( $M = 158.79$ ), complete technical or technological ( $M = 164.03$ ), complete professional ( $M = 167.77$ ), to postgraduate education ( $M = 176.99$ ). Students whose fathers were Ninguno scored slightly higher ( $M = 156.63$ ) than those whose fathers completed primary education ( $M = 155.20$ ). However, this discrepancy may be due to the small sample size of this category (only 123 students) and should not be considered as evidence that no education outperforms primary education.

A parallel ANOVA was conducted for mother's education using the same six categories. The result was also statistically significant,  $F(5, 9398) = 145.49, p < .001$ , indicating that the Global Score differed substantially across mother's education levels. Mean of the Global Score increased from Ninguno ( $M = 149.71$ ), complete primary education ( $M = 155.34$ ), complete secondary ( $M = 158.47$ ), complete technical or technological ( $M = 163.24$ ), complete professional ( $M = 168.85$ ), to postgraduate education ( $M = 175.64$ ). Unlike the father's education, no reversal was observed at the Ninguno category.

Both father's and mother's education showed strong associations with student academic outcomes. The two analyses produced nearly identical F-statistics, suggesting that father's and mother's education have comparable contributions and act as joint household resources. Students whose parents received professional or postgraduate

education scored higher than those whose parents did not. The higher the parents' education level, the higher the students' academic performance.

### **3.3 Socioeconomic Status and Performance Gaps**

The analysis examined whether socioeconomic status was associated with the Global Score using a one-way ANOVA. Stratum was recorded on a 1-6 scale, where higher values indicate higher socioeconomic status. The result was statistically significant,  $F(5, 12391) = 286.06$ ,  $p < .001$ , indicating substantial differences across strata. Mean of the Global Score increased from Stratum 1 ( $M = 152.35$ ), Stratum 2 ( $M = 158.20$ ), Stratum 3 ( $M = 163.82$ ), Stratum 4 ( $M = 172.00$ ), Stratum 5 ( $M = 177.39$ ), to Stratum 6 ( $M = 181.51$ ). With an F-statistic of 286.06, stratum emerged as the strongest predictor of student performance in this dataset.

SISBEN was also significantly associated with the Global Score,  $F(2, 4757) = 34.94$ ,  $p < .001$ . The analysis included three categories: Level 1 (the poorest), Level 2, and Level 3. Mean of the Global Score increased from Level 1 ( $M = 152.68$ ), Level 2 ( $M = 155.89$ ), Level 3 ( $M = 160.99$ ).

Both socioeconomic indicators pointed that students from more disadvantaged backgrounds scored lower on the Global Score. However, the magnitude of the gap differed between two measures. The stratum-based gap (29.16) was more than three times larger than the SISBEN-based gap (8.31). This difference might result from several factors. First, stratum was measured on a six-level scale, while SISBEN only had three categories. Second, the SISBEN sample was unevenly distributed, with Level 3 containing only 583 students compared to over 2,000 in Levels 1 and 2. Overall, socioeconomic disadvantage,

particularly as captured by stratum, is one of the strongest predictors of student performance in this dataset.

In addition, a closer examination of the stratum-based gap by subject module revealed that the largest disparity was in English proficiency (41.54 points between Stratum 1 and Stratum 6), which was more than twice the gap observed in any other module. This pattern suggested that English proficiency might be sensitive to socioeconomic background. It might be because of the unequal access to language learning resources outside the classroom (Pace et al. 292).

### **3.4 Gender Differences in the Global Assessment**

An independent-samples t-test showed a statistical difference in Global Scores between female and male students ( $M_{\text{female}} = 161.28$ ,  $M_{\text{male}} = 163.69$ ,  $t(12409) = 5.78$ ,  $p < .001$ ). However, the effect size was negligible (Cohen's  $d \approx 0.10$ ), indicating that the practical magnitude was small. The gender gap is much smaller than the socioeconomic disparities reported in Section 3.3 (29.16) and the parental education gap in Section 3.2 (25.93).

The pattern varied across subject modules. Female students outperformed male students in Written Communication ( $M_{\text{female}} = 56.62$ ,  $M_{\text{male}} = 51.71$ ). Male students outperformed female students in Quantitative Reasoning ( $M_{\text{female}} = 71.89$ ,  $M_{\text{male}} = 81.20$ ).

In summary, gender differences in academic performance were small, and was a much weaker predictor of student performance than socioeconomic status or parental education.

### **3.5 Limitations**

Several limitations should be noted. First, the dataset includes only Colombian engineering students, so the findings may not generalize to other disciplines or countries. Second, the result cannot establish causality, since the data is cross-sectional. Third, several variables contained unclassified or missing values (e.g. “Not sure” in parental education, STRATUM = 0, SISBEN = 0), which were excluded from the analyses, thus reducing the effective sample size.

#### **4. Recommendations**

First, it is recommended that a first-year bridging program be established for students from disadvantaged socioeconomic strata, particularly with the goal of improving English proficiency. The stratum-based gap was by far the largest in English (41.54), more than twice the gap observed in any other module.

Second, similar analyses can be conducted on students in other disciplines to determine whether the parental education and socioeconomic gaps observed here are specific to engineering students or represent a broader trend in higher education. If comparable disparities appear in other disciplines, the Department may need to consider interventions that beyond engineering programs.

#### **5. Conclusion**

This report analyzed the academic performance of 12,411 Colombian engineering students using data from *Data of Academic Performance Evolution for Engineering Students*. The analysis examined whether student performance was associated with parental education, socioeconomic stratum and gender.

The findings indicate that socioeconomic factors could be the strongest predictors of student performance. Students from Stratum 6 scored 29.16 higher than those from Stratum 1, representing the largest disparity observed in the analysis. Parental education showed comparably strong associations. Mean of the Global Score increased steadily with each successive level of parental education. However, gender did not have a substantial effect on student performance.

A closer examination of stratum-based gaps by module revealed that the disparity was largest in English, more than twice the gap observed in any other module. This finding suggests that improving English language support may be effective for narrowing the academic performance gap.

Overall, the results indicate that it is socioeconomic background and parental education that primarily impact academic disparities in this dataset, rather than gender.

## **6. References**

1. De La Hoz, Enrique. "Data of Academic Performance Evolution for Engineering Students." Mendeley Data, version 1, 2020, <https://doi.org/10.17632/83tcx8psxv.1>.
2. Pace, Amy, et al. "Identifying Pathways Between Socioeconomic Status and Language Development." *Annual Review of Linguistics*, vol. 3, 2017, pp. 285–308. <https://doi.org/10.1146/annurev-linguistics-011516-034226>.

## **7. Appendices**

### **Appendix A: Tables**

Table A1. Descriptive Statistics for Variables

| Variable | N      | MEAN   | SD    | MIN   | 25%    | 50%    | 75%    | MAX    |
|----------|--------|--------|-------|-------|--------|--------|--------|--------|
| QR_PRO   | 12,411 | 77.42  | 22.67 | 1.00  | 65.00  | 85.00  | 96.00  | 100.00 |
| CR_PRO   | 12,411 | 62.20  | 27.67 | 1.00  | 42.00  | 67.00  | 86.00  | 100.00 |
| CC_PRO   | 12,411 | 59.19  | 28.99 | 1.00  | 36.00  | 65.00  | 85.00  | 100.00 |
| ENG_PRO  | 12,411 | 67.50  | 25.50 | 1.00  | 51.00  | 74.00  | 88.00  | 100.00 |
| WC_PRO   | 12,411 | 53.70  | 30.00 | 0.00  | 28.00  | 56.00  | 80.00  | 100.00 |
| FEP_PRO  | 12,411 | 145.48 | 40.13 | 1.00  | 124.00 | 153.00 | 174.00 | 300.00 |
| G_SC     | 12,411 | 162.71 | 23.11 | 37.00 | 147.00 | 163.00 | 179.00 | 247.00 |

Table A2. Mean Global Score by Mother's Education Level

| Mother's Education Level               | N     | Mean G_SC |
|----------------------------------------|-------|-----------|
| Ninguno                                | 34    | 149.71    |
| Complete Primary                       | 713   | 155.34    |
| Complete Secondary                     | 3,106 | 158.47    |
| Complete Technical or<br>Technological | 1,495 | 163.24    |
| Complete Professional                  | 3,059 | 168.85    |
| Postgraduate                           | 997   | 175.64    |

Table A3. Mean of the Global Score by Father's Education Level

| Father's Education Level | N   | Mean G_SC |
|--------------------------|-----|-----------|
| Ninguno                  | 123 | 156.63    |
| Complete Primary         | 824 | 155.20    |

|                                        |       |        |
|----------------------------------------|-------|--------|
| Complete Secondary                     | 2,843 | 158.79 |
| Complete Technical or<br>Technological | 1,194 | 164.03 |
| Complete Professional                  | 3,016 | 167.77 |
| Postgraduate                           | 1,085 | 176.99 |

Table A4. Mean of the Global Score by Stratum

| Stratum             | N     | Mean G_SC |
|---------------------|-------|-----------|
| Stratum 1 (lowest)  | 1,709 | 152.35    |
| Stratum 2           | 4,029 | 158.20    |
| Stratum 3           | 4,045 | 163.82    |
| Stratum 4           | 1,578 | 172.00    |
| Stratum 5           | 633   | 177.39    |
| Stratum 6 (highest) | 403   | 181.51    |

Table A5. Mean of the Global Score by SISBEN

| SISBEN Level      | N     | Mean G_SC |
|-------------------|-------|-----------|
| Level 1 (poorest) | 2,057 | 152.68    |
| Level 2           | 2,120 | 155.89    |
| Level 3           | 583   | 160.99    |

Table A6. Stratum-Based Gaps by Subject Module

| Module | Stratum 1 | Stratum 6 | Gap |
|--------|-----------|-----------|-----|
|--------|-----------|-----------|-----|

|         |        |        |       |
|---------|--------|--------|-------|
| QR_PRO  | 70.52  | 88.36  | 17.84 |
| CR_PRO  | 53.57  | 73.76  | 20.20 |
| CC_PRO  | 50.92  | 71.07  | 20.16 |
| ENG_PRO | 50.91  | 92.45  | 41.54 |
| WC_PRO  | 49.30  | 65.59  | 16.28 |
| FEP_PRO | 142.98 | 162.17 | 19.19 |
| G_SC    | 152.35 | 181.51 | 29.16 |

Table A7. Mean Scores by Gender Across Subject Modules

| Module  | Female<br>(n = 5,043) | Male<br>(n = 7,368) | Gap<br>(F - M) | Cohen's d |
|---------|-----------------------|---------------------|----------------|-----------|
| QR_PRO  | 71.89                 | 81.20               | -9.31          | -0.42     |
| CR_PRO  | 61.34                 | 62.79               | -1.45          | -0.05     |
| CC_PRO  | 58.57                 | 59.61               | -1.04          | -0.04     |
| ENG_PRO | 66.53                 | 68.16               | -1.63          | -0.06     |
| WC_PRO  | 56.62                 | 51.71               | +4.91          | +0.16     |
| FEP_PRO | 145.30                | 145.60              | -0.30          | -0.01     |
| G_SC    | 161.28                | 163.69              | -2.41          | -0.11     |

## Appendix B: Python Code

### Code 1: Descriptive Statistics

```
scores_pro = ['QR_PRO', 'CR_PRO', 'CC_PRO', 'ENG_PRO', 'WC_PRO',
              'FEP_PRO', 'G_SC']
print(df[scores_pro].describe().round(2))
```

## Code 2: Parental Education ANOVA

```
print(df.groupby('EDU_FATHER')[scores_pro].mean().round(2))
print(df.groupby('EDU_MOTHER')[scores_pro].mean().round(2))

for name in df['EDU_FATHER'].unique():
    print(repr(name))

ef1 = df[df['EDU_FATHER']=='Ninguno']['G_SC']
ef2 = df[df['EDU_FATHER']=='Complete primary']['G_SC']
ef3 = df[df['EDU_FATHER']=='Complete Secundary']['G_SC']
ef4 = df[df['EDU_FATHER']=='Complete technique or technology']['G_SC']
ef5 = df[df['EDU_FATHER']=='Complete professional education']['G_SC']
ef6 = df[df['EDU_FATHER']=='Postgraduate education']['G_SC']

print("Ninguno:", len(ef1), ef1.mean())
print("Primary:", len(ef2), ef2.mean())
print("Secondary:", len(ef3), ef3.mean())
print("Technique or technology:", len(ef4), ef4.mean())
print("Professional education:", len(ef5), ef5.mean())
print("Postgraduate:", len(ef6), ef6.mean())

f,p = stats.f_oneway(ef1,ef2,ef3,ef4,ef5,ef6)
print("f-statistic:", round(f,3))
print("p-value:", p)

em1 = df[df['EDU_MOTHER']=='Ninguno']['G_SC']
em2 = df[df['EDU_MOTHER']=='Complete primary']['G_SC']
em3 = df[df['EDU_MOTHER']=='Complete Secundary']['G_SC']
em4 = df[df['EDU_MOTHER']=='Complete technique or technology']['G_SC']
em5 = df[df['EDU_MOTHER']=='Complete professional education']['G_SC']
em6 = df[df['EDU_MOTHER']=='Postgraduate education']['G_SC']

print("Ninguno:", len(em1), em1.mean())
print("Primary:", len(em2), em2.mean())
print("Secondary:", len(em3), em3.mean())
print("Technique or technology:", len(em4), em4.mean())
```

```

print("Professional education:", len(em5), em5.mean())
print("Postgraduate:", len(em6), em6.mean())

f,p = stats.f_oneway(em1,em2,em3,em4,em5,em6)
print("f-statistic:", round(f,3))
print("p-value:", p)

```

### Code 3: Stratum ANOVA

```

s1 = df[df['STRATUM']=='Stratum 1']['G_SC']
s2 = df[df['STRATUM']=='Stratum 2']['G_SC']
s3 = df[df['STRATUM']=='Stratum 3']['G_SC']
s4 = df[df['STRATUM']=='Stratum 4']['G_SC']
s5 = df[df['STRATUM']=='Stratum 5']['G_SC']
s6 = df[df['STRATUM']=='Stratum 6']['G_SC']

print("S1:", len(s1), s1.mean())
print("S2:", len(s2), s2.mean())
print("S3:", len(s3), s3.mean())
print("S4:", len(s4), s4.mean())
print("S5:", len(s5), s5.mean())
print("S6:", len(s6), s6.mean())

f,p = stats.f_oneway(s1,s2,s3,s4,s5,s6)
print("f-statistic:", round(f,3))
print("p-value:", p)

for score in scores_pro:
    s1 = df[df['STRATUM'] == 'Stratum 1'][score]
    s6 = df[df['STRATUM'] == 'Stratum 6'][score]
    print(f"{score}: Stratum 1 = {s1.mean():.2f}, Stratum 6 = {s6.mean():.2f}, gap = {s6.mean()-s1.mean():.2f}")

```

### Code 4: SISBEN ANOVA

```

for name in df['SISBEN'].unique():
    print(repr(name))

```

```

sis1 = df[df['SISBEN']=='Level 1']['G_SC']
sis2 = df[df['SISBEN']=='Level 2']['G_SC']
sis3 = df[df['SISBEN']=='Level 3']['G_SC']

print("Level 1:",len(sis1),round(sis1.mean(),2))
print("Level 2:",len(sis2),round(sis2.mean(),2))
print("Level 3:",len(sis3),round(sis3.mean(),2))

f,p = stats.f_oneway(sis1,sis2,sis3)
print("f-statistic:",round(f,3))
print("p-value:",p)

```

### Code 5: Gender t-test

```

print(df.groupby('GENDER')[scores_pro].mean().round(2))
male = df[df['GENDER'] == 'M']['G_SC']
female = df[df['GENDER'] == 'F']['G_SC']
t,p = stats.ttest_ind(male, female, equal_var = False)
print("Male mean:", round(male.mean(),2))
print("Female mean:", round(female.mean(),2))
print("Male SD:", round(male.std(),2))
print("Female SD:", round(female.std(),2))
print("Difference:", round(male.mean()-female.mean(),2))
print("t-statistic:",round(t,3))
print("p-value:", p)

```